

4. Operational Countermeasures: Homeostatic Resilience

This intelligence architecture transforms Artemis II data from a passive medical record into an active mission-readiness tool. When the GNN detects a Lévy Jump—an amber spike in the crew's collective stress graph—the system triggers the **Optimized Nutritional Support Window**.

It alerts Mission Control that the crew requires immediate homeostatic stabilization (e.g., an antioxidant, glucose, or B-vitamin payload) to prevent an impending cognitive or immune crash. This aligns directly with NASA's push toward autonomous crew health management for Mars-transit architectures.

5. Superlative Criterion Response

To further enhance the value of this existing data methodology, FTW Labs recommends the integration of **Continuous Real-Time Pupillometry (Eye-Tracking)** synchronized with **Localized Spacesuit Dosimetry**.

By correlating acute micro-doses of radiation with immediate changes in pupil dilation and saccade velocity, our sequence models could perfectly isolate radiation-induced neurological strain from standard physical EVA exhaustion. This would create the ultimate, high-fidelity precursor to deep-space autonomous health routing.